

Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus: Tyger Valley

Faculty: Information Technology

Module Code: ITPMA2-33

Group: 3

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Indicate	Yes	No
Plagiarism report attached		

Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I also declare that this assessment or any other of my original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. I am aware of the AI policy and acknowledge that I have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. I also declare that I am aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

Signature 	Date 28/08/2025
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Lecturer's Comments:

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Marks Awarded:	%
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Signature	Date
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Section A

1.1) Cost planning refers to the process of determining how costs should be managed throughout the life cycle of a project. This can be accompanied by appropriate policies, procedures and documentation that will be utilised when managing the project costs during the executing and controlling phases. Cost planning can be used to define information that will be utilised throughout the project such as what unit type should be used to measure project costs, like labour hours, how accurate the rounding guidelines for the project should be as well as how costs should be tracked to measure performance. This information can be altered in a manner that best suits a particular project based on the context, which allows the costs of the project to be managed effectively increasing the likelihood of the project's success in comparison to a project performed without these details.

Moreover, cost planning can also be used in conjunction with resource management to determine how the required resources of a project should be allocated to maximise its impact on the costs of the project, which can be crucial to a project's success. If the allocation of resources is performed in an efficient manner where the resources are used effectively, can save some project costs. However, an inefficient allocation of resources could lead to the project being delayed, which require more costs as the project will last for a longer period of time than was expected (Mohd Roshdi *et al.*, 2021).

Cost estimation refers to the process of estimating the potential cost of all resources required for the successful completion of a project. Cost estimation can be performed through multiple methods to ensure that the result is accurate for the specific project such as performing analogous estimates, which uses the costs from a similar project done before as a basis for the potential cost of the new project. Additional estimate techniques include bottom-up estimates, where a total is calculated by estimating and adding each individual work item and activity together, as well as parametric estimates, where a mathematical model is used to calculate the project's cost by using the project's characteristics as parameters.

Cost estimation is vital for projects to succeed, because decisions related to the project's budget, allocation of resources and cost baseline are influenced by the result of the cost estimation. An accurate cost estimation can greatly increase the effectiveness of the project's budget, ensure that resources are utilised as effectively as possible and create a realistic cost baseline that monitors the cost performance of the project. However, an inaccurate cost estimation can lead to various negative consequences including budget overruns, compromised project outcomes and delays, which could make the project a failure (SHAMIM MD, Mahfuzul.Islam., HAMID ABU BAKAR, Bin.Abdul., NYAMASVISVA, Tadiwa.Elisha. and RAFI NAJMUS, Saqib.Bin., 2025).

Cost controlling refers to the process of making any changes to the project and its budget in order to minimise costs where possible. Data from the cost performance can be analysed during cost controlling to determine potential risks in the project that can be used to predict outcomes and recognise the costs in vulnerable areas of the project (WULKE, Rita. and KOHL, B., 2004).

These identified risks can be notified to the project team and stakeholders, which can enable them to make appropriate changes to the project in order to minimise or remove these risks. These changes can reduce the size of the project's budget as new cost-effective methods are identified and utilised. This can increase the likelihood of the project succeeding, because these issues

could have resulted in the project team going over budget if they attempted to resolve it at a later date where the issue could become more damaging than previously.

Cost planning, estimation and controlling are linked to each other where each one provides a benefit to each other that can ensure that a project becomes successful. The information that is defined for a project in cost planning impacts how cost estimation and controlling are performed such as the rounding guideline being used to estimate costs and how performance is measured that determines if changes need to be made for the project's costs. Defining this information is beneficial as it ensures that all aspects of project cost management behave in a manner that is most suitable for the specific project. Cost estimation also can be used in conjunction with cost controlling to provide an additional area for a project manager to monitor and determine if changes need to be made to ensure project success. Moreover, cost controlling could be used to alter the original plan in cost planning if the change is required to ensure project success, which can impact itself as well as cost estimation.

1.2) An agile project management framework refers to a flexible approach where changes can be easily made throughout the project, however it lacks a structure needed for large projects. The agile project management framework utilises various methods throughout the project's lifecycle such as scrum, an iterative and incremental approach to agile software project management where communication between stakeholders and project team members is encouraged and specific tasks are complete in specified time periods known as sprints. Moreover, additional agile methods include kanban, a method that can be used in conjunction with scrum and is used to visualise the workflow, define process policies and improve collaborative efforts, as well as extreme programming, a method that promotes an iterative approach and enables for essential practices to be integrated rapidly for an immediate outcome (Santoso, Raharjo and Wibowo, 2025).

A waterfall project management framework refers to a plan-driven approach that provides a predictable and coherent structure that a project team can follow to ensure that large-scale projects are controllable and documented, however it can't easily implement changes during the project's life cycle as it lacks the flexibility that the agile project management framework offers (Masih, 2025). The waterfall project management framework strictly follows sequential stages consisting of a project analysis, design, construction, testing and support.

In terms of scope management, the agile project management framework is effective for projects where aspects of the scope is unclear. This framework enables the project team to receive feedback from the stakeholders and make the appropriate changes throughout the life cycle of the project to ensure that any misconceptions in the project's scope are resolved. However, the flexibility offered by the agile project management framework could also be a disadvantage for a project, because frequent changes to the scope can lead to scope creep and prolonged product delivery times (Saunders, 2018).

The waterfall project management framework contrasts the agile framework where any misunderstandings regarding the scope of the project can't be easily resolved and could lead to a project failure. Therefore, the waterfall framework should only strictly follow the scope of projects where there is little to no uncertainty in the content of the scope and it is not diverse (HASSAN, Ali., YOUNAS, Soayba. and BHAUMIK, Amiya., 2020). The waterfall project management framework is effective for these types of projects on a larger scale, because the coherent structure of this framework and documentation ensures that new project team members are able to understand the project's scope in full detail. It also enables project team members that can only work with the rest

of the project team in a virtual team environment to be on the same page as the rest of the project team despite being in a different time zone.

In terms of budget management, the agile project management framework can provide a more flexible budget than the waterfall project management framework that can easily accommodate for any changes. The flexibility of the agile project management framework enables the project team to make essential changes to the project in a manner that does not require an extensive amount of funds from the project's budget, unlike the waterfall project management framework where changes could make the project go over budget as changes can't be made as easily as the agile project management framework. However, determining and managing the budget of a project that uses the agile project management framework during the early stages of its lifecycle can be quite difficult, because drastic changes could occur that could require the project manager to change the project's budget to fit the drastic changes.

The structured approach that the waterfall project management framework offers can enable the project manager to determine a more accurate budget for the project than the agile project management framework. The waterfall project management framework is used for projects where the scope is widely understood. The budget will be comprised of each aspect of the project that requires funds, such as equipment and expertise, is identified and documented. This enables the project manager to manage the budget of the project more effectively than the agile project management framework as well as identify and resolve areas in the project where the project could go over budget.

In terms of timeline control, the agile project management framework can make use of scrum to create multiple deadlines of 2 – 4 weeks called a sprint. Each sprint is made up of specific objectives of a project, that was defined by the product owner in the product backlog, that the project team will have to complete during the life-cycle of the sprint. After each sprint is concluded, a new sprint with other objectives from the product backlog is created and started by the project team. This format of timeline control enables the project manager, also known as the ScrumMaster, to easily manage smaller sections of the project over a small period of time. Dividing the project into sprints enables the deadline to be more flexible than a fixed schedule from the waterfall project management framework to accommodate anything that could require the deadline to be pushed back.

However, the waterfall project management framework makes use of a fixed schedule instead of scrum, because this approach provides a clear roadmap that outlines the projects objectives. The project manager can utilise the roadmap to predict the most appropriate fixed schedule for the project where all identified project objectives can be completed within the predicted deadline. Fixed schedules that are identified due to the project's predictability makes the waterfall project management framework effective for timeliness projects (Masih, 2025).

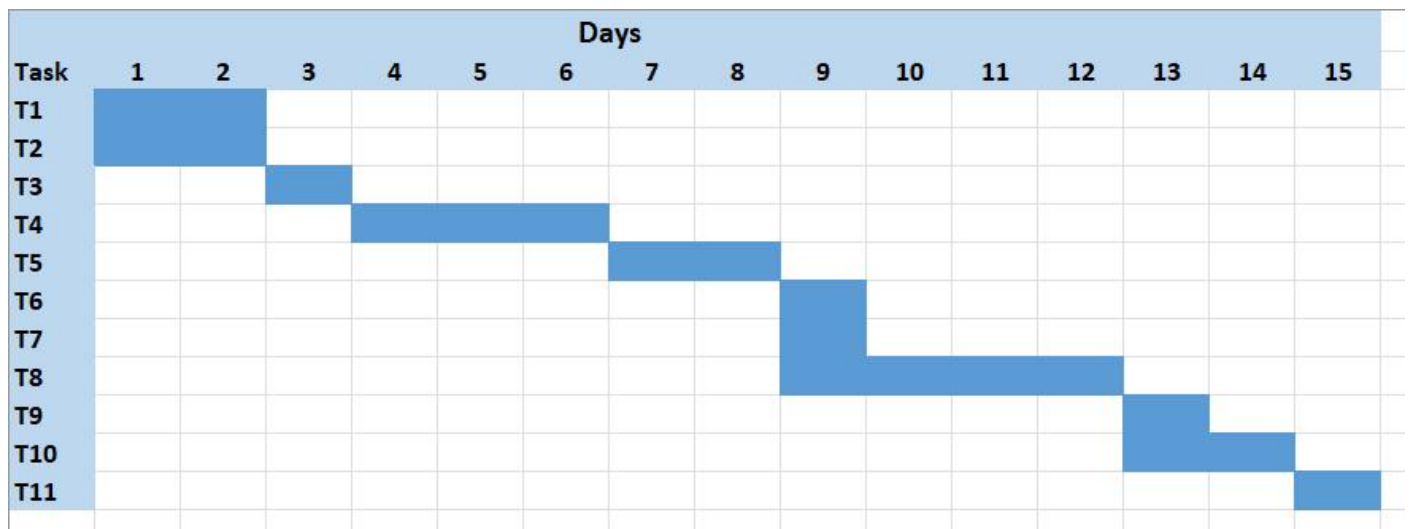
1.3) The project manager should hold a meeting with the project team members where the rescheduling requested by the project sponsor should be discussed and a new schedule is created. Project team members can assist the project manager with creating the new schedule by identifying the remaining tasks of the project that should be implemented in the new schedule to ensure that efficiency and project stability are considered. (GE, Yujia. and XU, Bin., 2016).

After the remaining tasks are identified, the project manager should then determine the relationships between each task. Identifying these relationships provides the project manager with a better understanding of the aspects of each task, enabling them to make decisions on the new schedule that ensures that it meets the deadline.

The project manager can then utilise the tasks and it's relationships to define how long and when certain tasks should be done to ensure that each task can be accomplished realistically by the project team. An appropriate time allocation for each task should enable the project team members to complete tasks in a manner where the quality is acceptable and each dependent task is not delayed due to specific tasks not being completed.

Moreover, the project manager should utilise the agile methodology that the project team is already making use of to effectively monitor the new schedule and make necessary changes. These changes should be conducted to the new schedule if any inefficiencies are identified, when the project manager monitors the progress of the new schedule, that could result in the new deadline being missed.

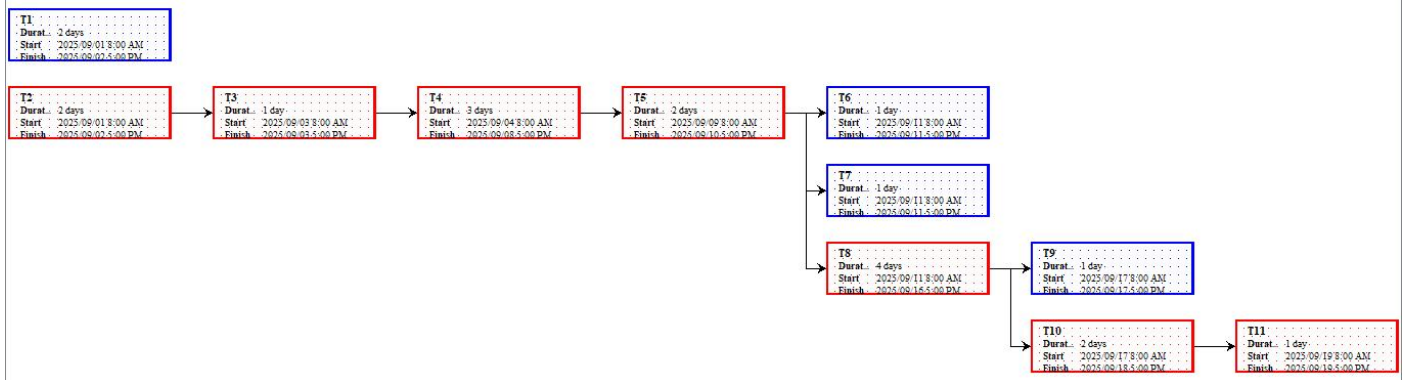
1.4)



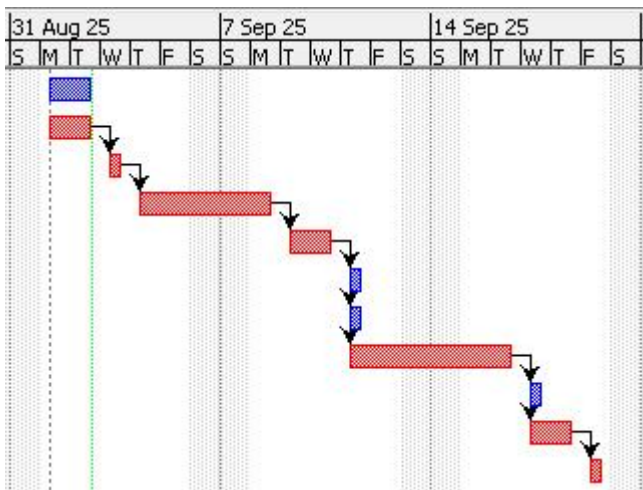
1.5)

Task dependency table:

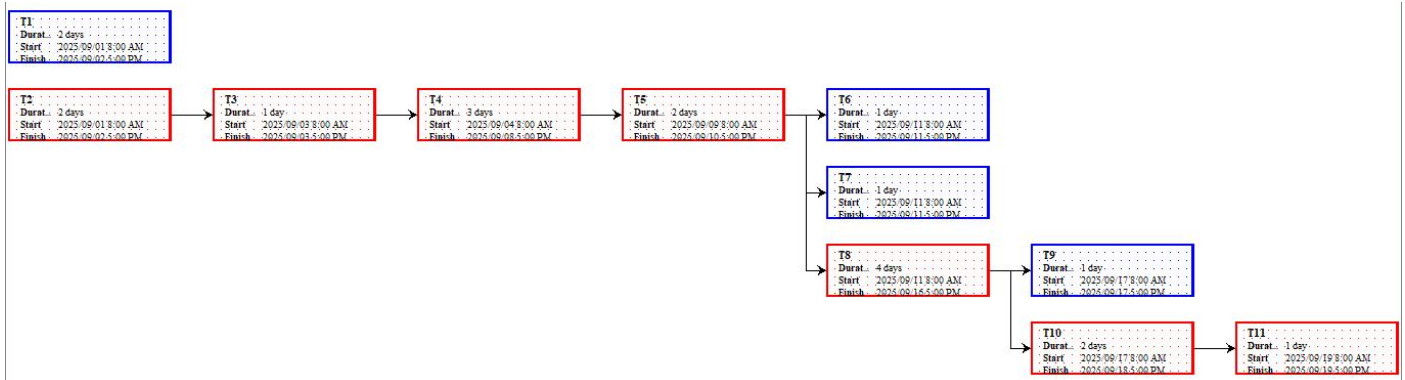
	Name	Duration	Start	Finish	Predecessors
1		2 days	2025/09/01 8:00 AM	2025/09/02 5:00 PM	
2		2 days	2025/09/01 8:00 AM	2025/09/02 5:00 PM	
3		1 day	2025/09/03 8:00 AM	2025/09/03 5:00 PM	2
4		3 days	2025/09/04 8:00 AM	2025/09/08 5:00 PM	3
5		2 days	2025/09/09 8:00 AM	2025/09/10 5:00 PM	4
6		1 day	2025/09/11 8:00 AM	2025/09/11 5:00 PM	5
7		1 day	2025/09/11 8:00 AM	2025/09/11 5:00 PM	5
8		4 days	2025/09/11 8:00 AM	2025/09/16 5:00 PM	5
9		1 day	2025/09/17 8:00 AM	2025/09/17 5:00 PM	8
10		2 days	2025/09/17 8:00 AM	2025/09/18 5:00 PM	8
11		1 day	2025/09/19 8:00 AM	2025/09/19 5:00 PM	10



Gantt chart:



Network diagram:



1.6)

Path A: $T1(2) = 2$

Path B: $T2(2) + T3(1) + T4(3) + T5(2) + T6(1) = 9$

Path C: $T2(2) + T3(1) + T4(3) + T5(2) + T7(1) = 9$

Path D: $T2(2) + T3(1) + T4(3) + T5(2) + T8(4) + T9(1) = 13$

Path E: $T2(2) + T3(1) + T4(3) + T5(2) + T8(4) + T10(2) + T11(1) = 15$

Path E is the critical path.

Float:

Path A : $CP(15) - 2 = 13$ extra days

Path B : $CP(15) - 9 = 6$ extra days

Path C : $CP(15) - 9 = 6$ extra days

Path D : $CP(15) - 13 = 2$ extra days

The critical path analysis refers to a project management technique that is used to display all tasks and it's relationships in a project. It assists project managers to ensure that the project can be completed within a timeframe and allows the project manager to reorganise if required (LEE, Sangkon., 2019). This is accomplished through the critical path method that divides a project into individual tasks where the project manager can identify the route in the project with the longest duration that determines the total time spent on the project, which is referred to as the critical path (Huang and Zhang, 2025). The critical path can by the project manager to determine what tasks they should monitor for any potential issues. An issue that could occur includes one or more tasks that unexpectantly takes longer to complete, where the project manager could take action and make changes to ensure that the project schedule doesn't slip.

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